Criterion A: Problem Specification

Defining the Problem

The goal is to create a software solution that is able to accurately visualize and simulate the interaction between rigid bodies. These interactions should be visualized from beginning to end, with the user being able to stop the simulation at any time, and make any changes to the simulation. These changes can include moving, scaling, or accelerating any of the objects in the simulation, or adding and deleting objects from the simulation. In addition, the user will be able to see any of the information in the simulation at all times. This can include the velocity of objects, their current position relative to the center of the simulation, and their size and mass. The visualization of these interactions can help better understand many more complicated physics problems relating to forces of mechanics.

Rationale for Proposed Solution

The solution will be coded using the Processing language. This is a Java based language, with simplified syntax, as well as graphics and UI capabilities. A key component of processing is its P3D renderer, which this project will be using, as the simulation will be a 3-dimensional simulation. The simulation will be 3-dimensional, to allow for the greatest degree of freedom for what the simulation can run. The ability to change, add, or delete objects will also allow the user to set up whatever situation or problem they want.

Success Criteria

1. The simulation will accurately represent the physical interaction between the objects
2. The user will have the ability to add and remove objects
3. The user will be able to modify existing objects